

PROJECT DOCUMENT

BIOMASS ENERGY & AI

Reforestation in Mitsue

御杖村プロジェクト概要 / Mitsue Village Project — Summary

御

Version	v3.4
Date	2026-07-08
Author	Rob Oudendijk

御杖村プロジェクト概要 / Mitsue Village Project — Summary

One-Page Briefing for Mitsue Village Government

What We Propose

To repurpose the **former Sugano Elementary School (Mitsue Taiken Koryukan / 御杖体験交流館)** — an available community exchange center and the leading candidate site (final choice confirmed in Phase 1) — as a **community sustainability center** that will:

1. Restore aging cedar (sugi) plantations to healthy native forest — native broadleaf trees feed local wildlife, reducing crop raids by deer, wild boar, and bear; thinning the sugi also generates biomass fuel for the village energy system
2. Generate the village's electricity and heat from sugi forest thinnings through a biomass CHP system — a circular energy economy where the forest powers the village, complemented by solar, EV charging, and community backup power (battery storage subject to feasibility study)
3. House a small, low-impact data center powered by that local energy — a working model of progress and sustainability as a single coherent system

A 25-year initiative built carefully, in close partnership with the village.

This Project Delivers Your Own Renewable Energy Plan

The BIOMASS ENERGY & AI project is the **implementation and operating vehicle (官民連携 運営体制)** for the village's official "**Plan for Maximum Introduction of Renewable Energy**" (御杖村再エネ導入最大化計画, Jan 2025) — the MoE-funded strategy the village has already written and committed to. We do not ask the village to adopt a new idea; we offer to execute the plan it already has.

- **The "one resilient site" target.** Your plan sets a target of **one** renewable + storage + EV resilient site in the village by 2045 — currently **zero**. Our data center, with on-site biomass CHP generation and battery backup, can credibly **be** that site, letting the village report progress against its own published indicator.
- **EV charging priority.** Your plan makes EV chargers at public facilities (town hall, roadside stations) a **priority**, because transport is **46% of village emissions** — the largest source. Our EV charging network anchored to local generation delivers exactly this.
- **Distributed disaster-resilience.** Your plan calls for an autonomous, distributed energy society that can power public buildings and clinics during outages. Our 24/7 biomass CHP + solar + battery provides this.
- **Forest carbon / J-Credit mechanism — uniquely ours.** Your plan wants forest CO₂ absorption (0.3 → **2.7 kt/yr** by 2045) and a J-Credit system, but has no operator or offtaker for the thinnings. Our biomass CHP **is** that mechanism: it consumes sugi thinnings, funding faster thinning and native restoration, producing the certified forest carbon the plan calls for.
- **Funding that flows through the village.** Because the village completed the MoE planning grant (step 1), it is eligible for the **地域脱炭素移行・再エネ推進交付金** (step 2): a **2/3–3/4 subsidy** on solar/battery/EV/private-wire capex (Mitsue qualifies as a 過疎 + low-financial-capacity area). The grant is paid **to the village** via public-private partnership — the village brings the plan and the grant channel; we bring the operator, capital, and anchor offtaker.

The village's published figures: 9 kt-CO₂ emissions, 46% transport, ~90% forest cover, 60% reduction by 2030, carbon-neutral by 2045. Source: 御杖村再エネ導入最大化計画 (概要版), Jan 2025.

Direct Benefits to Mitsue Village

- **Local employment** — forestry work, facility operations, maintenance roles
- **Income for landowners** — fair compensation to private mountain owners for sugi harvesting
- **Lower energy costs** — locally generated biomass and solar power for the village
- **Circular forest-to-energy economy** — sugi thinnings fuel the village's biomass CHP system, turning a forest liability into local energy
- **EV charging infrastructure** — Mitsue prepared for the coming shift to electric vehicles, powered by the local energy system
- **New use for the former Sugano school (Koryukan)** — preserving and honoring an important community facility
- **Reduced cedar pollen** — gradual replacement of sugi with native species over decades
- **Reduced wildlife crop damage** — restored native forest provides natural food for deer, wild boar, and bear, keeping them out of agricultural areas
- **Broadband and digital infrastructure** — improved connectivity for residents and businesses
- **National and international visibility** — Mitsue as a model village for sustainable rural revitalization

Early Benefits — Visible Within Five Years

The 25-year horizon applies to forest restoration ecology. Concrete benefits to the village arrive much earlier:

Year	Tangible benefit to the village
1	Former Sugano school (Koryukan) reactivated as datacenter hub; village named in domestic and international press; first researcher and student visits
2	First forestry contracts → direct cash income to private mountain owners
2–3	Reduced sugi pollen and landslide risk in actively managed plots
3–4	EV charging stations operational; battery storage if confirmed by feasibility study
4–5	Data center hires local staff; hosting fees and energy sales become real revenue; village becomes a study-tour destination
Ongoing	Community programmes, field trips, and open environmental and financial dashboards from Year 1 onward

Economic Value to the Village (Indicative)

Precise figures will come from Phase 1 feasibility studies, but early-stage estimates suggest meaningful local economic value:

- **Energy currently imported:** approximately ¥40–60M per year leaves Mitsue's local economy as electricity payments. Local generation keeps a portion of this value within the village.
- **Former Sugano school maintenance:** currently ¥3–8M per year in upkeep with no return. Active use generates value from a previously stranded asset.
- **Direct employment:** an estimated 8–15 local jobs by Year 5 (forestry, facility operations, EV charging, administration).
- **Landowner income:** new revenue streams for private mountain owners through sustainable cedar harvesting.
- **Combined annual economic activity** by Year 5 (estimated): ¥28–67M, with a meaningful share retained locally.

These ranges are illustrative; the formal numbers will be developed with the village in Phase 1.

What We Are Asking For

At this early stage, we are seeking:

- An **opportunity to discuss** the project with village leadership
- Guidance on how to engage the **community (自治会) and residents** appropriately
- A **letter of interest** to support our application for initial feasibility study funding

We are **not** asking for village funds at this stage.

How the Project Will Be Organized

A dedicated **non-profit organization (NPO法人)** will be established to coordinate the project. The NPO will work transparently with:

- Mitsue Village Government
- Local residents and community associations (自治会)
- Private forest landowners
- Forestry contractors
- Nara Prefecture and national agencies (林野庁, METI)
- Academic and technical partners

Who We Are

Rob Oudendijk — Dutch electrical engineer born in the Netherlands (the same country that is home to ASML, where innovation is natural), Mitsue resident since 2012 (Sugano), core hardware developer for Safecast (the global citizen science radiation monitoring network).

San Poisson — Project Manager

Advisors:

- **Ray Ozzie** (Executive Chair, Blues)
- **Takuo Dome** (Professor, Osaka University Graduate School of Economics; Representative Director, Inochi Forum)
- **Evin Zoet** (Co-Representative Director, Transom)
- **Yoshiko Zoet-Suzuki** (Co-Representative Director, Transom)

Our Approach

- **Local first.** Mitsue residents and landowners are the foundation of this project.
- **No surprises.** We will share information openly and seek guidance at every step.
- **Patient and modest.** We are building for 25 years, not for headlines.
- **Replicable.** What we learn in Mitsue will be shared so other villages can benefit.

Next Step

We respectfully request a meeting with village leadership at their convenience to discuss the project and listen to your guidance.

Rob Oudendijk Sugano, Mitsue Village, Nara Prefecture [Contact details]